DTM-17: Running a Cluster Health Assessment

> **Series:** DTM — Dynatrace Managed | **Reference:** 17 — Running a Cluster Health Assessment | **Created:** October 2026 | **Last Updated:** 11/11/2026
>
> **Dynatrace Classic only.** This series applies to **Dynatrace Managed**, the self-hosted, on-premises deployment. None of it applies to Dynatrace SaaS.

Overview

DTM-99 is the routine a cluster's own team runs every day, week and month. This notebook is for a different occasion: a **one-off review** of a Managed cluster, by someone from outside the team or by the team itself before a decision. Typical triggers are a hardware purchase, a licence renewal, a move to Managed-to-SaaS, a new site, or a run of incidents nobody can explain.

A review like that needs things the routine doesn't: a way to rate each check, a rule for what to do when the evidence is missing, a list of what to ask for before starting, and a summary someone else can act on. This notebook supplies those, and an ID'd checklist that points each line back to the notebook holding its detail and sources.

What you'll find here:

- How an assessment differs from the routine checks, and why it is read-only
- A four-value rating scale, and the rule against guessing
- Where each kind of evidence comes from, and what to request up front
- A checklist of nine areas with stable IDs (T1, D8, L2 …), each mapped to the five health layers
- How to roll the findings up into something a cluster owner can act on

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Prerequisites

Requirement	Details
Dynatrace deployment	A self-hosted **Dynatrace Managed** cluster. Nothing in this series applies to Dynatrace SaaS.
Access	Cluster Management Console (CMC) access for the reviewer; root (or `sudo`) on the nodes for whoever runs the node commands; a cluster API token with `ServiceProviderAPI` for the node-list call
Query language	None. Managed has no Grail, so this series has **no DQL cells**
Read first	DTM-10 (the five-layer health model) and DTM-99 (the routine checklist this one extends)
Version basis	Written against the Managed documentation read 10/29/2026

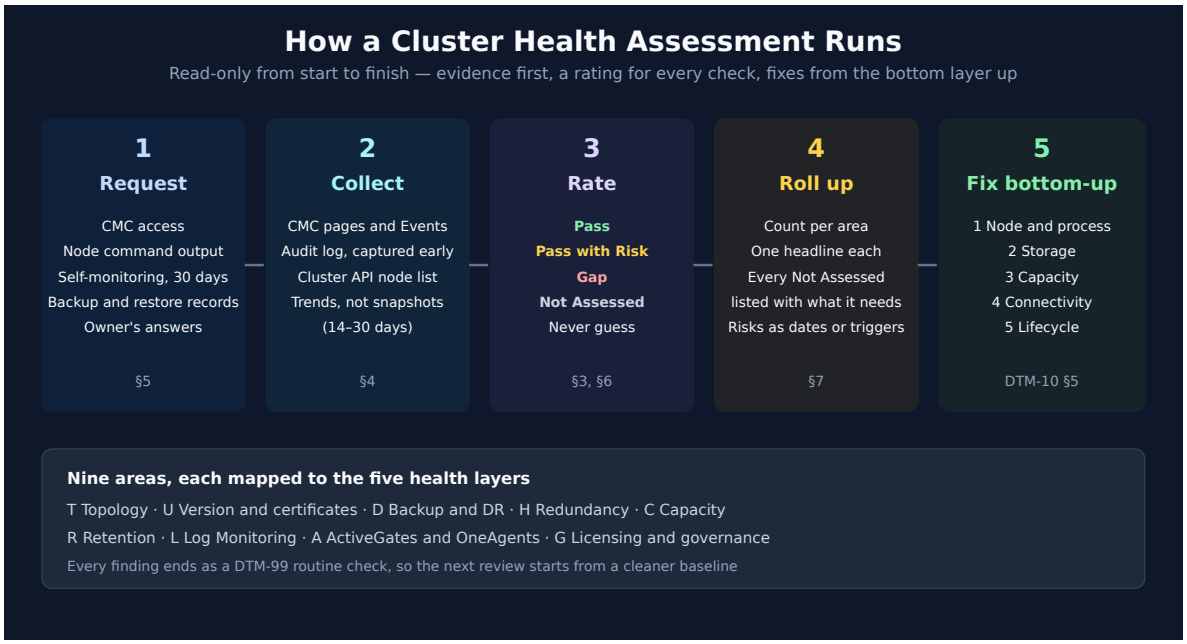
1. Short Answer

1. **Work read-only.** Every check below reads a CMC page, an event list, an API response, a node command's output or a document. None of them changes the cluster.
2. **Rate every check:** *Pass*, *Pass with Risk*, *Gap*, or *Not Assessed*. Use **Not Assessed** whenever the evidence is missing, and say what would settle it. Never fill a gap with a guess.
3. **Trend, don't snapshot.** Capacity and disk checks need weeks of history. One reading taken on a quiet afternoon proves very little.
4. **Ask for the evidence before you start.** CMC access, node command output, self-monitoring access, the backup and restore records, and the owner's answers on recovery objectives take longest to arrive.
5. **Roll up by area and fix bottom-up.** A count per area, one headline each, and every *Not Assessed* listed with what it needs. Fix the lowest failing layer first (DTM-10 §5).

2. Assessment or Routine?

	Routine checks (DTM-99)	Assessment (this notebook)
Who runs it	The cluster's own operators	A reviewer, often from outside the team, with the owner answering questions
When	Daily, weekly, monthly, after every change	Once, before a decision or after a run of incidents
Access	Full administrator and root	As little as the review needs; node commands run by the owner and shared
Output	A fixed problem, or nothing	A rated checklist and a summary someone else acts on
Missing evidence	Go and look	Rate it *Not Assessed* and list what is needed
Design questions	Assumed settled	Re-opened: is the availability model, the retention, the recovery plan still right?

The two share most of their checks. An assessment adds the questions the routine assumes are settled: the availability model, recovery objectives, a second cluster if there is one, and whether the sizing still matches the load.



3. Rating Each Check

Rating	Meaning
Pass	Evidence is present and within the healthy range
Pass with Risk	Within range today, but a trend, a peak, an unowned item or an unverified assumption puts it at risk
Gap	Outside the healthy range, or a required capability is missing
Not Assessed	The evidence wasn't available in the review window, or needs the owner's input. Name what is needed

Three rules keep the ratings honest:

- **Not Assessed is a finding, not a failure.** It tells the owner what to produce. Hiding it, or turning it into a *Pass* because nothing looked wrong, is how a backup that has never been restored ends up rated healthy.
- **A rating rests on evidence you can point to.** A screenshot, a command's output, an event list or the owner's written answer. A verbal "yes, that's fine" moves a check from *Not Assessed* to *Pass with Risk*, not to *Pass*.
- **Read the threshold off the current documentation page** before quoting it in a report. The values in §6 come from the notebooks they point to, which record the date they were read. Limits change between releases.

The rating scale and these rules are this series' method. Dynatrace doesn't publish an assessment method for Managed clusters (§8).

4. Evidence Sources

Source	What it gives you	Where the detail is
CMC	*Home* (including the last backup), *Deployment status* (nodes, ActiveGates, network zones), *Environments*, *Events*, *Settings*, *Licensing*, *Audit log*	DTM-10 §6.1
CMC Events	Every cluster event, including the ones that never email	DTM-99 §4
CMC Audit log	*"Audit log lists events triggered by the cluster and selected configuration changes for all environments."* Only *"changes related to cluster-wide configuration, licensing, storage quota, and permissions"*; environment-level entries come from the Audit Logs API	DTM-10 §6.1
CMC environment export	Each environment's page has an **Export configuration** action: *"To export the configuration of the selected Dynatrace environment"*. Useful for comparing two clusters (D8)	DTM-16 §8.1
Cluster API v1	`onpremise/cluster`: every node's `operationState` and build version; `onpremise/cluster/configuration`: roles and data-center assignment	DTM-10 §6.4, DTM-11 §2
Node commands	`dynatrace.sh status` / `check`, `cassandra-nodetool.sh status`, `_cluster/health`, `du` per store. They need root, so in a read-only review the owner runs them and shares the output	DTM-11 §3, DTM-12 §5, DTM-13 §5
Self-monitoring	Local self-monitoring (`dsfm:` metrics and the capacity indicator), hosted, or private, depending on licensing. Private self-monitoring adds host CPU, memory and disk from OneAgents on the nodes	DTM-10 §6.3, DTM-15 §5.3
Mission Control	Dynatrace's view of the cluster. The owner sees connection status in the CMC; for anything more, ask Dynatrace	DTM-10 §6.5, DTM-14 §6
Diagnostic archive	Per-node logs and configuration, for questions the CMC can't answer	DTM-11 §9
The owner	Recovery objectives, restore records, the off-site copy, planned changes, the reason behind any held update	—
Documentation	The thresholds, read on the day of the review	The notebook each row in §6 points to

Which self-monitoring you can use depends on licensing (DTM-10 §6.3). Local self-monitoring is documented as DDU-only on one page and as on by default on another. Hosted self-monitoring is Dynatrace's, and private self-monitoring consumes licences. Find out which one exists before you plan a check that needs it.

> **Sources:** Cluster Management Console (DT docs) — *"For all environment audit log entries, use the Audit Logs API."*

5. What to Request Before You Start

The checks that stall a review are almost always waiting on evidence. Ask for it before the review window opens:

1. **CMC access** for the reviewer, and a Mission Control remote-access grant if Dynatrace will take part, covering the whole review window (DTM-14 §8).
2. **Self-monitoring access**, or exports from it, covering at least the last 30 days.
3. **Node command output** from every node: `dynatrace.sh status` and `check`, `cassandra-nodetool.sh status`, `_cluster/health`, and `du` for each store. If a question needs more, a diagnostic archive from the affected node.
4. **The node-list API response** (`onpremise/cluster` and `onpremise/cluster/configuration`).
5. **The backup and recovery file:** the restore runbook, the date and result of the last rehearsal, where the off-site copy is, and the recovery objectives the business has agreed.
6. **The change calendar** for the review window: patches, upgrades, reboots, hardware changes.
7. **Planned changes:** new log volume, new sites, new retention requirements, new environments.
8. **For a second cluster:** the same items for it, and an *Export configuration* of each environment on both sides.

In community practice, a review window of two to four weeks fits most clusters: long enough to collect trends and wait for answers, short enough that the CMC Events and Audit log still cover it. Verify the window against your own cluster's event history.

6. The Assessment Checklist

Each area maps to one or more of the five layers from DTM-10 (1 node and process, 2 storage, 3 capacity, 4 connectivity, 5 lifecycle) or to the design and configuration review in DTM-99 §3. **Healthy** gives the documented value where there is one. The **Detail** column names the notebook that quotes the source. Rows marked **(owner)** need the owner's input and are rated *Not Assessed* until it arrives.

6.1 T — Topology and design (layer 1, design review)

ID	Check	Evidence	Healthy	Detail
T1	Node count and hardware	CMC *Deployment status*; `onpremise/cluster`	At least three nodes, at most 42 (Managed 1.348 and later; 30 before), every node the same hardware configuration	DTM-01 §3, DTM-15 §3
T2	Node size against its class	CMC node detail; the sizing table	Each node meets its class minimum; at least 64 GB RAM on every node if Log Monitoring is on	DTM-15 §2, DTM-13 §3
T3	Availability model **(owner)**	`onpremise/cluster/configuration` (data centers); the owner	Chosen and written down: single cluster, rack awareness, Premium HA, or a second cluster, each with the failure it covers	DTM-01 §4
T4	Premium HA prerequisites (if used)	CMC; the owner	Three or more nodes per data center, both sized the same, round trip at most 100 ms, online	DTM-16 §7.1

ID	Check	Evidence	Healthy	Detail
T5	Latency, time and operating system	Node output; host monitoring	Latency between nodes 10 ms or less; NTP and one time zone; every node on the same supported OS version	DTM-01 §3, §6; DTM-11 §7, §8

6.2 U — Version, updates and certificates (layers 4 and 5)

ID	Check	Evidence	Healthy	Detail
U1	Installed version	`onpremise/cluster` (`buildVersion`); the release-notes table	The same build on every node, inside "Currently supported"	DTM-11 §2, DTM-16 §6
U2	Update mode and holds **(owner)**	CMC *Settings* > *Automatic update*; the owner	Automatic updates in a maintenance window, or a manual routine that keeps up; any hold has an owner and a reason	DTM-16 §5
U3	Install partition	Node `df` output	At least 5 GB free on the partition where Managed is installed	DTM-16 §5.2
U4	Certificates **(owner)**	CMC node detail; the owner's certificate inventory	Expiry tracked outside the cluster, with a named owner. The expiry events never email	DTM-14 §5
U5	Mission Control link	CMC *Licensing* > *Check Mission Control connection*	Connected	DTM-14 §6, §7
U6	Remote access	CMC *Settings* > *Remote access permissions*	Scope and role chosen deliberately; review access limited to the review window	DTM-02 §4, DTM-14 §8

6.3 D — Backup and disaster recovery (layer 5)

ID	Check	Evidence	Healthy	Detail
D1	Backup enabled, scope chosen	CMC *Settings* > *Backup*	Enabled; user sessions and log events included or excluded on purpose	DTM-16 §2, §3
D2	Backups succeeding	CMC *Home* (last backup); Events	A recent last backup; no *Cassandra backup problem* or *ElasticSearch backup problem* events in the window	DTM-16 §3
D3	Scope understood **(owner)**	The owner	The owner knows transaction storage is never backed up and recent 1- and 5-minute metrics are not either	DTM-16 §2
D4	Backup target	Node mounts; CMC *Settings* > *Backup*	Shared storage at the same path on every node, NFSv4 recommended, mounted persistently, with space for the backup size estimate	DTM-16 §3
D5	Off-site copy **(owner)**	The owner	A copy kept off-site	DTM-16 §3

ID	Check	Evidence	Healthy	Detail
D6	Restore readiness **(owner)**	The restore kit and runbook	Installer version, node inventory, seed choice and matching UID:GID ready; a rehearsal on record	DTM-16 §4
D7	Recovery objectives **(owner)**	The owner	RTO and RPO agreed per data type, and compared with what a backup restore gives (the docs' comparison table: *RPO = 24h; RTO = 24h*)	DTM-16 §2
D8	Second cluster (if any) **(owner)**	Both CMCs; *Export configuration* from each environment on both sides	Configuration kept in step, retention long enough, a documented way for agents and users to reach it, and the capacity to take the load	DTM-16 §8.1

6.4 H — Redundancy and failure tolerance (layers 1 and 2)

ID	Check	Evidence	Healthy	Detail
H1	Copy counts understood	The documentation	Three copies of metrics, configuration and user sessions; two of log events; transaction storage distributed, not replicated	DTM-10 §4
H2	Failures the plan assumes	The owner's plan	Plans for one lost node. The docs keep three copies at every size; a Dynatrace blog allows two from five nodes	DTM-10 §4
H3	Capacity with one node gone	Self-monitoring trend; the arithmetic	Survivors stay below full load after one loss; on three nodes that means typical load at or below about two-thirds	DTM-15 §4.1
H4	Outage history	CMC Events; Audit log	Few node or host outages, each with a known cause	DTM-11 §6
H5	Cassandra	`cassandra-nodetool.sh status`	Every node `UN`; `Load` roughly even	DTM-12 §5
H6	Elasticsearch	`_cluster/health`	`green`	DTM-13 §5
H7	Cluster ActiveGates	CMC *Deployment status* > *ActiveGates*	None *Unknown*; none flagged as more than five versions behind	DTM-14 §9

6.5 C — Capacity and sizing (layers 2 and 3)

ID	Check	Evidence	Healthy	Detail
C1	Load against headroom	Self-monitoring, 14–30 days	Typical load leaves one-third headroom	DTM-15 §4, §5.3
C2	Adaptive load reduction	CMC Events	None, or brief; never 15 minutes or more	DTM-15 §5.1

ID	Check	Evidence	Healthy	Detail
C3	Memory pressure	CMC Events	No heap memory emergency mode or hard cleanup events	DTM-15 §5.2, DTM-11 §7
C4	Cluster traffic control	CMC Events	None. It stops monitoring new hosts, and it never emails	DTM-15 §5.2
C5	Metrics store size	`du` per node	Below 2 TB per node, and even across nodes	DTM-12 §3
C6	Disk space	CMC Events; `du` per store	No *Insufficient disk space* events; Elasticsearch sizes even across nodes; growth trended	DTM-12 §6, DTM-13 §5.2
C7	Load against the node class	CMC licensing figures; the sizing table	Host units and Full-Stack OneAgents within the per-node limits, whichever is reached first	DTM-15 §2
C8	Ingest limit after hardware changes **(owner)**	CMC *Environments* > *Cluster overload prevention*; change calendar	Reviewed after the last hardware change	DTM-15 §6
C9	Throughput	`dsfm:server.*` self-monitoring metrics	Steady or growing; no unexplained drop	DTM-15 §5.3

6.6 R — Retention (layer 2)

ID	Check	Evidence	Healthy	Detail
R1	Retention per environment **(owner)**	CMC *Environments* > *Storage settings*	Every user-session retention above the 35-day default, and every custom log retention, is a decision someone owns	DTM-13 §4
R2	Retention being cut	CMC Events	No *Transaction storage retention period truncation*: adaptive retention isn't trimming history	DTM-10 §7
R3	Quotas	CMC *Environments*	Set on purpose; none hit unexpectedly	DTM-03 §3, DTM-15 §6

6.7 L — Log Monitoring (layers 2 and 3; skip if not used)

ID	Check	Evidence	Healthy	Detail
L1	Hardware for logs	CMC node detail	64 GB RAM or more on every node	DTM-13 §3
L2	Rejected log events	`dsfm:server.log_and_events_monitoring.events_rejected_count`	Zero, or rarely above zero	DTM-13 §6

ID	Check	Evidence	Healthy	Detail
L3	Incoming volume	`dsfm:server.log_and_events_monitoring.events_incoming_count`	Trend known; peaks known	DTM-13 §6
L4	Log pressure	CMC Events	No *Ingested log data is trimmed*, *Log ingest queue is full*, *Elasticsearch log queue is full* or *Elasticsearch log storing failed*	DTM-13 §6
L5	Two copies understood **(owner)**	The owner	The owner knows a second node failure makes some log events unavailable	DTM-13 §2
L6	Volume forecast **(owner)**	L3 and the owner's plans	Planned volume converted to daily storage and compared with the store's size (below)	DTM-13 §3
L7	Logs in the backup **(owner)**	CMC *Settings* > *Backup*	Included or excluded on purpose	DTM-13 §8

Converting a log rate to daily volume. Planned log volume usually arrives as events per minute, while storage is sized in bytes. Measure the average event size rather than assuming it, then:

> GB per day ≈ events per minute × average bytes per event × 1,440 ÷ 1,000,000,000

For example, 50,000 events a minute at 600 bytes each is about 43 GB a day before copies. Multiply by two for the two copies Log Monitoring keeps (DTM-13 §2).

6.8 A — ActiveGates and OneAgents (layer 4, design review)

ID	Check	Evidence	Healthy	Detail
A1	ActiveGate roles	CMC *Deployment status* > *ActiveGates*	Each ActiveGate has one purpose; Environment ActiveGates next to the OneAgents they serve	DTM-04 §2, §3
A2	ActiveGate redundancy and load	CMC; host monitoring of the ActiveGate hosts	Two or more per data center per job; CPU at or below 50% and memory at or below 80%, so the others can take over	DTM-04 §4
A3	ActiveGate updates	CMC	Environment ActiveGates on a set update mode; Cluster ActiveGates within five versions	DTM-04 §7
A4	ActiveGate tokens	CMC Events	No *ActiveGate Token(s) will expire soon.* left unhandled	DTM-14 §9
A5	Network zones	CMC *Deployment status* > *Network zones*	Zones follow the network layout; a fallback mode chosen for each	DTM-04 §5, DTM-05 §4
A6	OneAgent versions and policy	CMC; environment settings	Update policy set by host group; versions within support; hosts below the x86-64-v2 baseline identified	DTM-05 §5, §6
A7	OneAgent coverage **(owner)**	The owner's host inventory	Monitored hosts match the expected inventory	—

In community practice, reviewers take the agent total from one place, not by adding up the per-ActiveGate connection counts. Those counts can overlap. Verify against your own CMC before reporting a number.

6.9 G — Licensing and governance (design review)

ID	Check	Evidence	Healthy	Detail
G1	Licence consumption and expiry **(owner)**	CMC *Licensing*; the owner	Consumption below the licence at peak, not only today; expiry date known	—
G2	Licensing and the Mission Control link	CMC *Licensing*	On classic licensing, no outage near 14 days	DTM-14 §7
G3	Events read	CMC Events, the whole window	Every warning and severe event explained; none of the never-emailed events left unread	DTM-99 §4
G4	Audit log	CMC *Audit log*; Audit Logs API per environment	Cluster-wide configuration, licensing, storage-quota and permission changes in the window explained	§4
G5	Notifications reach people	CMC *Settings* > *Emails*; the notification settings object	SMTP working, recipients and emergency contacts set, configurable events set explicitly	DTM-02 §3, DTM-10 §6.2
G6	Sign-in and access	CMC *User authentication*	LDAP or SSO; the installation administrator kept as a break-glass account; a small cluster-admin group	DTM-02 §6, DTM-03 §4–§6

ID	Check	Evidence	Healthy	Detail
G7	Host hardening **(owner)**	The owner	Nodes patched to one level and hardened to the organisation's standard	DTM-01 §6

> **Sources:** Cluster Management Console (DT docs), Self-monitoring metrics (DT docs), Pre-upgrade information (DT docs). Every other value is quoted, with its source, in the notebook named in the Detail column. **Derived:** the daily-volume formula is unit arithmetic; the doubling applies the two-copy rule in DTM-13 §2.

7. Rolling Up the Findings

A checklist of nearly sixty lines isn't a result. Roll it up so the owner can see where to start:

Area	Pass	Pass with Risk	Gap	Not Assessed	Headline
T — Topology and design					
U — Version, updates, certificates					
D — Backup and disaster recovery					
H — Redundancy					
C — Capacity					
R — Retention					
L — Log Monitoring					
A — ActiveGates and OneAgents					
G — Licensing and governance					

Then:

1. **List every *Not Assessed* check with what would settle it.** This is the owner's to-do list, and often the most useful page of the report.
2. **Order the gaps bottom-up** by the DTM-10 layers: node and process, storage, capacity, connectivity, lifecycle. A storage gap can surface as a capacity symptom, so fix the lower layer first.
3. **Write each *Pass with Risk* as a date or a trigger.** "Metrics store reaches 2 TB on node 2 in about four months at the current rate" can be acted on. "Disk is getting full" can't.
4. **Hand the routine back.** Most findings turn into a DTM-99 check that keeps them from coming back. Point to the line.

8. What the Documentation Does Not Say

Not found, in the Managed documentation read 10/29/2026:

Gap	Working assumption in this notebook
A Dynatrace method for assessing a Managed cluster: ratings, evidence, a scored result	The rating scale in §3 and the roll-up in §7 are this series' own
`dsfm:` self-monitoring metrics for Server garbage collection, queue overflows or sampling	Use the events (C2–C4) and, with private self-monitoring, the host and process metrics from the nodes' OneAgents
Where the per-environment log ingest limit is set, and how it relates to any cluster-wide figure	Read the rejected-events metric (L2). The Log Monitoring troubleshooting page answers *"Who can increase the ingestion limit per tenant?"* by pointing to a Dynatrace product expert
What *Export configuration* contains, and in what format	Export both sides and compare (D8); don't assume it covers every setting
How long the CMC Audit log and Events are kept	Capture both at the start of the review window

Thresholds this checklist leaves to judgment. The documentation gives no value for these. In community practice, reviewers use the following, so state them as practice in a report rather than as limits, and verify them against your own cluster:

Check	Value used in community practice
Disk on a data volume (C6)	Warn at 85% used. It is the lowest of Elasticsearch's upstream watermarks (DTM-13 §5.2)
Certificate expiry (U4)	Renewal planned 90 days out; a *Gap* within 30 days
Restore rehearsal (D6)	Within the last 12 months
Licence expiry (G1)	More than six months away

> **Sources:** Self-monitoring metrics (DT docs), Log Monitoring troubleshooting (DT docs), Cluster Management Console (DT docs), Manage your monitoring environments (DT docs). **Observed 10/29/2026:** the self-monitoring metrics page lists no garbage-collection, queue or sampling metric; the environments page describes log quotas and Log Monitoring storage, but no per-minute ingest limit; the CMC page gives no retention for the Audit log or Events.

9. Recommended Approach

1. **Send the request list (§5) a week before the review starts**, and agree who runs the node commands.
2. **Capture CMC Events and the Audit log on day one**, before anything ages out.
3. **Rate every check in §6**, with the evidence beside it. When the evidence is missing, rate *Not Assessed* and write down what is needed.
4. **Ask the owner the (owner) questions in one sitting**, in writing. Recovery objectives (D7) and the second cluster (D8) take the longest to answer.
5. **Roll up by area (§7)**, order the gaps bottom-up, and turn each *Pass with Risk* into a date or a trigger.

6. **Hand the cluster back to the routine.** Each finding should end as a DTM-99 check, so the next review starts from a cleaner baseline.

10. Summary and Next Steps

A cluster health assessment is DTM-99's checklist run once, by a reviewer, with three additions: a rating for every check, an honest *Not Assessed* where evidence is missing, and the design questions the routine takes as settled. Ask for the evidence early, trend rather than snapshot, roll the results up by area, and fix from the bottom layer up.

Next in the series:

Notebook	Use it for
DTM-99: Best Practice Summary and Health-Check Checklist	The routine checks a finding should turn into
DTM-10: Managed Cluster Architecture and Health Model	The five layers and where health is surfaced
DTM-16: Backup, Upgrade, and Disaster Recovery	Recovery objectives, restore readiness and a second cluster

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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.